42 Scientific Reliability of Virtual Reconstructions: Uncertainty Assessment and Geometric Validation of Algeria's Archaeological Heritage — The Cases of Timgad and Djemila

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Timgad and Djemila are important archaeological sites, representing Roman remains that demonstrate advanced urbanism and architectural understanding. The digital reconstruction process opens pathways for interpreting missing architectural elements; however, the lack of complete archaeological records makes it difficult to determine which reconstructed models may be distorted and to distinguish what is real (fact) from what is invented (hypothetical).

In this article, a framework is proposed for the assessment of hypothetical 3D representations of Timgad and Djemila, based on two approaches, uncertainty assessment and geometric verification. The first one focuses on identifying the evidence behind reconstructed features, recording assumptions made during reconstruction and presenting the degree of confidence using a system of classifications and metadata. The second approach includes analysis of dimensional consistency, as well as geometric and topological correctness, depending on the quality and availability of archaeological records and survey data.

Additionally, the framework pays attention to semantic organization, provenance documentation, interoperability and reproducibility which are essential components of the validation of research findings. FAIR principles are particularly important for ensuring the findability, accessibility, interoperability and reusability of reconstruction models and their supporting materials. The comparison of different reconstruction hypotheses may provide additional clarity in the on the relationship between archaeological data and interpretations made by specialists.

By connecting digital reconstruction in conjunction with the preservation and interpretation of Algeria's archaeological heritage, this study considers 3D models as hypotheses open to revision rather than definitive representations of the past. Through this approach, it seeks to contribute to more reliable methodologies for architectural and archaeological reconstruction.